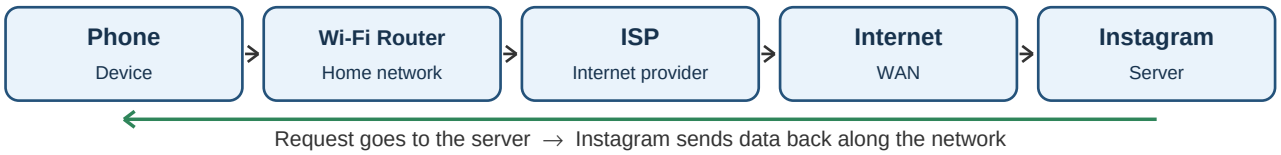
Networking Assignment

Name: _____ Class: _____

1. Diagram: How My Phone Connects to Instagram

When I open Instagram, my phone sends a request through the Wi-Fi router and Internet Service Provider (ISP). The request travels across the Internet to Instagram's server. The server sends the requested posts, photos, or videos back to my phone.

How a phone connects to Instagram



Connections: Phone to router = Wi-Fi (WLAN); router to ISP = broadband connection; ISP to Instagram = Internet/WAN. The reply returns through the network to the phone.

2. Five Devices and Their Network Types

Device	Network type	How it connects
Smartphone	WLAN / WAN	Uses Wi-Fi at home or mobile data outside.
Laptop	LAN / WLAN	Connects to a home or college network by cable or Wi-Fi.
Smart TV	LAN / WLAN	Connects to the router to stream videos.
Smartwatch	PAN	Often connects to a phone using Bluetooth.
Wi-Fi router	LAN / WAN	Connects local devices and links them to the ISP/Internet.

Note: LAN means Local Area Network, WLAN means Wireless LAN, WAN means Wide Area Network, and PAN means Personal Area Network.

3. Difference Between the Internet and an Intranet

Internet: The Internet is a worldwide public network. People can use it to visit websites, watch videos, send messages, and search for information. For example, I can use public Wi-Fi to open Google or Instagram.

Intranet: An intranet is a private network used by an organization, such as a college or company. For example, a college may have a private portal where students can check attendance, notices, or study materials. Only authorized users can access it.

Main difference: The Internet is public and worldwide, while an intranet is private and limited to an organization.

4. How Data Travels When I Search in WhatsApp

1. I open WhatsApp on my phone and type a name or keyword in Search.

2. My phone processes the search and, when an online request is needed, sends data through Wi-Fi or mobile data.

3. The data travels through my router or mobile network and ISP to WhatsApp's servers over the Internet.

4. The server processes the request and sends the required response back through the network to my phone.

5. WhatsApp displays the matching results. Some searches may happen directly on the phone, depending on the type of search.

Conclusion

Networks allow devices to communicate and share information. The Internet connects networks around the world, while LANs, WLANs, and PANs connect devices in smaller areas.